MIL-M-38510/172B
30 April 1984
SUPERSEDING
MIL-M-38510/172A
29 October 1979

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, CMOS, AND-OR-INVERT, EXCLUSIVE-OR, EXCLUSIVE-NOR GATES, MONOLITHIC SILICON, POSITIVE LOGIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, CMOS logic microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Dual 2-wide, 2-input AND-OR-invert gate
02	Expandable 4-wide, 2-input AND-OR invert gate
03	Quad exclusive-OR gate
04	Quad exclusive-NOR gate

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat package
X	F-1 (14-lead, 1/4" x 1/4"), flat package, except A dimension = 0.1" (2.54 mm) max.
Y	F-2 (14-lead, 1/4" x 3/8"), flat package, except A dimension = 0.1" (2.54 mm) max.

NOTES

1. As an exception to nickel plate or undercoating paragraph of MIL-M-38510, for case outlines X and Y only, the leads of bottom brazed ceramic packages (i.e. configuration 2 of case outlines F-1 or F-2) may have electroless nickel undercoating which shall be 50 to 200 microinches (1.27 to 5.08 µm) thick provided the lead finish is hot solder dip (i.e. finish letter A) and provided that, after any lead forming, an additional hot solder dip coating is applied which shall extend from the outer tip of the lead to no more than 0.015 inch (0.38 mm) from the package edge.
2. For bottom or side-brazed packages, case outlines X and Y only, the S_1 dimension may go to .000 inch (.00 mm) minimum.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: National Aeronautics and Space Administration, George C. Marshall Space Flight Center, ATTN: EG02, Marshall Space Flight Center, AL 35812, using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Absolute maximum ratings.

Supply voltage range (V_{DD} - V_{SS})	-0.5 V to +18 V
Input current (each input)	±10 mA
Input voltage range	(V_{SS} - 0.5) $\leq V_I \leq$ (V_{DD} + 0.5)
Storage temperature range	-65°C to +175°C
Maximum power dissipation (P_D)	200 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case	(see MIL-M-38510, Appendix C)
Junction temperature (T_J)	+175°C

1.4 Recommended operating conditions.

Supply voltage(V_{DD} - V_{SS})	4.5 V dc to 15 V dc
Input low (V_{IL}) voltage range	0 - 1.5 V dc at V_{DD} = 5 V dc, V_{OL} = 10% V_{DD}, V_{OH} = 90% V_{DD}; 0 - 4.0 V dc @ V_{DD} = 15 V dc, 0-2.0 V dc @ V_{DD} = 10 V dc
Input high (V_{IH}) voltage range	3.5 - 5.0 V dc at V_{DD} = 5 V dc, V_{OL} = 10% V_{DD}, V_{OH} = 90% V_{DD}; 11.0 - 15.0 V dc @ V_{DD} = 15 V dc, 8.0 - 10.0 V dc @ V_{DD} = 10 V dc.
Ambient operating temperature range (T_A)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein. Although eutectic die bonding is preferred, epoxy die bonding may be performed. However, the resin used shall be Dupont 5504 Conductive Silver Paste, or equivalent, which is cured at 200°C ±10°C for a minimum of 2 hours. The use of equivalent epoxies or cure cycles shall be approved by the qualifying activity. Equivalency shall be demonstrated in data submitted to the qualifying activity for verification.

3.2.1 Logic diagram and terminal connections. The logic diagrams and terminal connections shall be as specified on figure 1.

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3.2.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.3 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained and available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III. V_{IL} and V_{IH} testing requires only a summary of attributes data.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.6.1 Total dose radiation hardness identifier. Total dose radiation hardness identifier shall be in accordance with MIL-M-38510 and 4.5.5 herein.

3.6.2 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.3 Correctness of indexing and markings. All devices shall be subjected to the final electrical tests specified in table II after part number marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 36 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Delete the sequence specified in 3.1.9 through 3.1.13 of method 5004 and substitute lines 1 through 7 of table II herein.

b. Burn-in (method 1015 of MIL-STD-883).
   (1) Static tests (test condition A) use circuit shown on figure 3, or equivalent. Ambient temperature (T_A) shall be 125°C minimum. Test duration for each static test shall be 24 hours minimum for class S devices and in accordance with table I of method 1015 for class B devices.

   (2) Dynamic test (test condition D) using circuit shown on figure 4, or equivalent. Ambient temperature (T_A) shall be 125°C minimum. Test duration shall be in accordance with table I of method 1015.

c. Interim and final electrical test parameters shall be as specified in table II, herein.

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d. For class S devices, post dynamic burn-in, or class B devices, post static burn-in, electrical parameter measurements may, at the manufacturer's option, be performed separately or included in the final electrical parameter measurements.

4.2.1 Percent defective allowable (PDA).

a. The PDA for class S devices shall be 5 percent for static burn-in and 5 percent for dynamic burn-in, based on the exact number of devices submitted to each separate burn-in.

b. Static burn-in I and II failures shall be cumulative for determining PDA.

c. The class B devices PDA shall be in accordance with MIL-M-38510 for static burn-in. Dynamic burn-in is not required.

d. Those devices whose measured characteristics after burn-in exceed the specified delta (Δ) limits or electrical parameter limits specified in table III, subgroup 1, are defective and shall be removed from the lot. The verified failures divided by the total number of devices in lot initially submitted to burn-in shall be used to determine the percent defective for the lot and the lot shall be accepted or rejected based on the specified PDA.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 5, 6, 7, and 8 of table I of method 5005 of MIL-STD-883 shall be omitted.

c. Subgroup 4 (C_I measurement) shall be measured only for initial qualification and after process or design changes which may affect input capacitance. Capacitance shall be measured between the designated terminal and V_{SS} at a frequency of 1 MHz.

d. Subgroup 12 shall be added to the group A inspection requirements for class S devices, using an LTPD of 15, and consist of the test conditions and limits specified in table III.

e. At the manufacturer's option (with the exception of device type 02), test tapes may be programmed simultaneously for each identical section, provided that each output is measured and each specified input combination is tested.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Class S steady state life (accelerated) test circuits shall be submitted to the qualifying activity for approval. When the alternate steady state life test is used, the circuit on figure 4, or equivalent, shall be used.

b. A special subgroup shall be added using an LTPD of 15 for classes S and B, and measured only for initial qualification and after process or design changes. This subgroup shall consist of a high voltage test of the input protection circuits, V_{ZAP} (see 4.5.3).

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions V_{SS}=0 V -55°C ≤ T_A ≤ 125°C Unless otherwise specified		Device type	Limits Min	Limits Max	Unit
Positive clamping input to V_{DD}	$V_{IC(pos)}$	T_A=25°C, V_{DD}=GND, V_{SS}=Open, Output=Open I_{IN} = -1 mA		All	-	1.5	Vdc
Negative clamping input to V_{SS}	$V_{IC(neg)}$	T_A=25°C, V_{DD}=Open, V_{SS}=GND, Output=Open I_{IN}=1 mA		All	-	-6.0	Vdc
Quiescent supply current	I_{SS} or I_{DD}	V_{DD}=18 Vdc, V_{IN}=V_{SS} or V_{DD} All input combinations		All	-	-750	nA
High-level output voltage	V_{OH1}	V_{DD}=15 Vdc Any one input=V_{DD} All other inputs=V_{SS} $\|I_O\|$ ≤ 1 µA		01,02,03	14.95	-	Vdc
		V_{DD}=15 Vdc V_{IN}=V_{SS} V_{IN}=V_{DD} $\|I_O\|$ ≤ 1 µA		04			
Low-level output voltage	V_{OL1}	V_{DD}=15 Vdc V_{IN}=V_{DD} $\|I_O\|$ ≤ 1 µA		01,02	-	0.05	Vdc
		V_{DD}=15 Vdc V_{IN}=V_{DD} V_{IN}=V_{SS} $\|I_O\|$ ≤ 1 µA		03			
		V_{DD}=15 Vdc Any one input=V_{DD} All other inputs=V_{SS} $\|I_O\|$ ≤ 1 µA		04			
Input high voltage	V_{IH1}	V_{DD}=5 Vdc Any one input=3.5 Vdc All other inputs=V_{SS} $\|I_O\|$ ≤ 1 µA	V_O=4.5 Vdc V_O=0.5 Vdc V_O=4.5 Vdc V_O=0.5 Vdc	01 02 03 04	3.5		Vdc
	V_{IH2}	V_{DD} = 10 Vdc Any one input=7.0 Vdc All other inputs=V_{SS} $\|I_O\|$ ≤ 1 µA	V_O=9.0 Vdc V_O=1.0 Vdc V_O=9.0 Vdc V_O=1.0 Vdc	01 02 03 04	7.0		Vdc
	V_{IH3}	V_{DD}=15 Vdc Any one input=11.0 V All other inputs=V_{SS} $\|I_O\|$ ≤ 1 µA	V_O=13.5 Vdc V_O=1.5 Vdc V_O=13.5 Vdc V_O=1.5 Vdc	01 02 03 04	11.0		Vdc

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions V_{SS}=0 V -55°C ≤ T_A ≤ 125°C Unless otherwise specified		Device type	Limits Min	Limits Max	Unit
Input low voltage	V_{IL1}	V_{DD}=5 Vdc All other inputs=V_{SS} Any one input=1.5 V V_O=0.5 Vdc $\|I_O\|$ ≤ 1 µA	V_O= 4.5 Vdc	01,04		1.5	Vdc
			V_O= 0.5 Vdc	02,03			
	V_{IL2}	V_{DD}=10 Vdc All other inputs=V_{SS} Any one input=3.0 V V_O = 1.0 Vdc $\|I_O\|$ ≤ 1 µA	V_O= 9.0 Vdc	01,04		3.0	Vdc
			V_O= 1.0 Vdc	02,03			
	V_{IL3}	V_{DD}=15 Vdc All other inputs=V_{SS} Any one input=4.0 V V_O=1.5 Vdc $\|I_O\|$ ≤ 1µA	V_O= 13.5 Vdc	01,04		4.0	Vdc
			V_O= 1.5 Vdc	02,03			
Output high (source) current	I_{OH1}	V_{DD}=5 Vdc V_{IN}=V_{SS} V_{OH}=4.6 Vdc		01 02 04	-0.36		mAdc
		V_{DD}=5 Vdc Any one input=V_{DD} All other inputs=V_{SS} V_{OH}=4.6 Vdc		03			
	I_{OH2}	V_{DD}=15 Vdc V_{IN}=V_{SS} V_{OH}=13.5 Vdc		01 02 04	-2.4		mAdc
		V_{DD}=15 Vdc Any one input=V_{SS} All other inputs=V_{SS} V_{OH}=13.5 Vdc		03			
Output low (sink) current	I_{OL1}	V_{DD}=5 Vdc V_{IN}=V_{SS} V_{OL}=0.4 Vdc		01 02 03	0.36		mAdc
		V_{DD}=5 Vdc Any one input=V_{DD} All other inputs=V_{SS} V_{OL}=0.4 Vdc		04			
	I_{OL2}	V_{DD}=15 Vdc V_{IN}=V_{SS} V_{OL}=1.5 Vdc		01 02 03	2.4		mAdc
		V_{DD}=15 Vdc Any one input=V_{DD} All other inputs=V_{SS} V_{OL}=1.5 Vdc		04			

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TABLE I. Electrical performance characteristics-Continued.

Test	Symbol	Conditions V_{SS}=0 V -55°C ≤ T_A ≤ 125°C Unless otherwise specified	Device type	Limits		Unit
				Min	Max	
Input leakage 1/ current	I_{IH}	V_{DD}=18 Vdc Measure inputs sequentially; connect all unused inputs to V_{SS}	All		45	nA
Input leakage 1/ current	I_{IL}	V_{DD}=18 Vdc Measure inputs sequentially; connect all unused inputs to V_{DD}	All		-45	nA
Input capacitance	C_I	V_{DD}=0 Vdc, f=1 MHz, T_A=25°C	All		7.5	pF
Input test voltage	V_{ZAP}	C1=100 pF R2=1.5 kΩ (see 4.5.3)	All	400		V
Propagation delay times High to low level	t_{PHL}	V_{DD}=5 Vdc, C_L=50 pF, R_L=200 kΩ (see figure 5).	01,02	15	630	ns
			03,04	13	350	ns
Low to high level	t_{PHL}		01	25	870	ns
			02	25	980	ns
			03,04	13	350	ns
Transition times High to low level	t_{THL}	V_{DD}=5 Vdc, C_L=50 pF, R_L=200 kΩ (see figure 5)	All	10	280	ns
Low to high level	t_{TLH}		All	10	280	ns
Propagation delay High to low level	t_{PHL}	V_{DD}=10 Vdc, C_L=50 pF, R_L=200 kΩ, T_A=25°C (see figure 5)	01,02	6	180	ns
			03,04	6	120	ns
Low to high	t_{PLH}		01	10	250	ns
			02	14	280	ns
			03,04	6	120	ns
Transition times High to low level	t_{THL}	V_{DD}=10 Vdc, C_L=50 pF, R_L=200 kΩ, T_A=25°C (see figure 5)	All	5	100	ns
Low to high	t_{TLL}		All	5	100	ns

1/ Input current at one input node.

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TABLE II. Electrical test requirements.

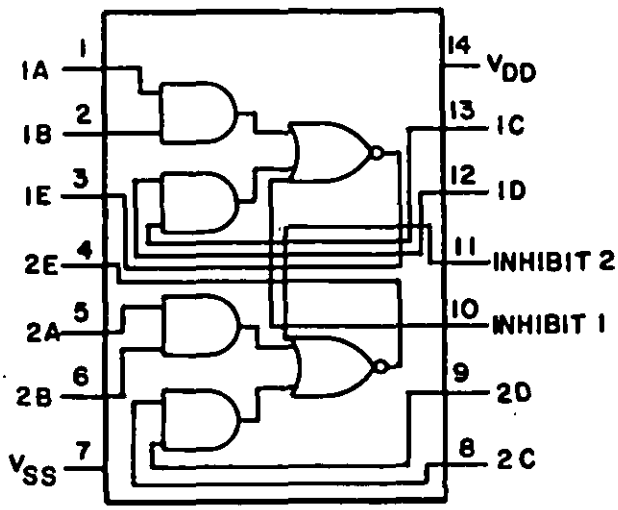
Line no.	MIL-STD-883 test requirements	Class S device 1/				Class B device 1/			
		Ref. par.	Table 2/ III subgroups	Table 3/ IV delta limits	Test circuit figure	Ref. par.	Table 2/ III subgroups	Table 3/ IV delta limits	Test circuit figure
1	Interim electrical parameters method 5004		1				1		
2	Static burn-in I method 1015	4.2b 4.5.2			3				
3	Same as line 1		1	▲					
4	Static burn-in II method 1015	4.2b 4.5.2			3	4.2b 4.5.2	4/		3
5	Same as line 1		1*	▲		4.2b	1*	▲	
6	Dynamic burn-in method 1015	4.2b 4.5.2			4				
7	Same as line 1	4.2b	1*	▲					
8	Final electrical parameters method 5004		1*2,3,9				1*2,3,9		
9	Group A test requirements method 5005	4.4.1	1,2,3,4,9 10,11,12			4.4.1	1,2,3,4,9		
10	Group B end point electrical parameters method 5005	4.4.2	1,2,3,9, 10,11	▲					
11	Group C end point electrical parameters method 5005					4.4.3	1,2,3	▲	
12	Additional group C tests method 5005					4.4.3c	10,11		
13	Group D end point electrical parameters tests method 5005	4.4.4	1,2,3			4.4.4	1,2,3		

1/ Blank spaces indicate tests are not applicable.
2/ (*) indicates PDA applies to subgroup 1 (see 4.2.1).
3/ (▲) indicates delta limit shall be required only on table III, subgroup 1, where specified, and delta values shall be computed with reference to the previous interim electrical parameters.
4/ The device manufacturer may, at his option, either perform delta measurements or within 24 hours after burn-in (or removal of bias) perform the final electrical parameter measurements.

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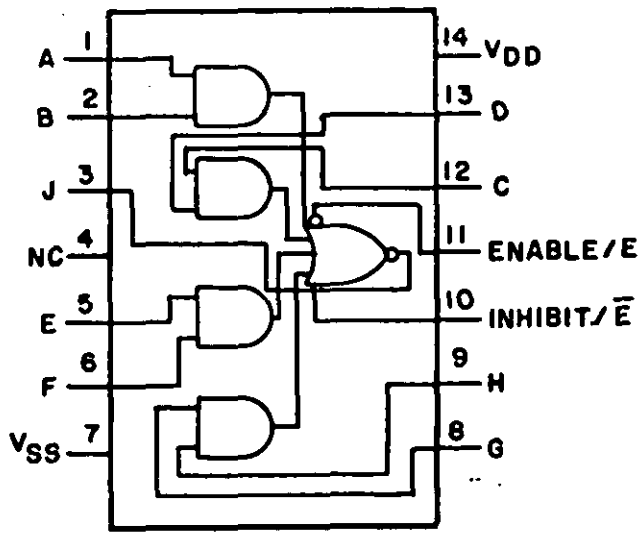
Device type 01

Cases A, C, D, X, and Y



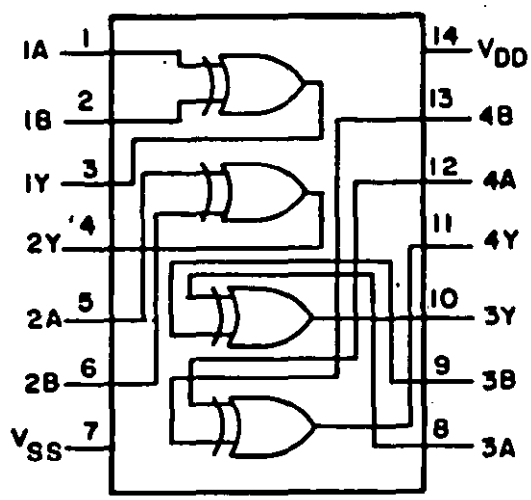
Device type 02

Cases A, C, D, X, and Y



Device type 03

Cases A, C, D, X, and Y



Device type 04

Cases A, C, D, X, and Y

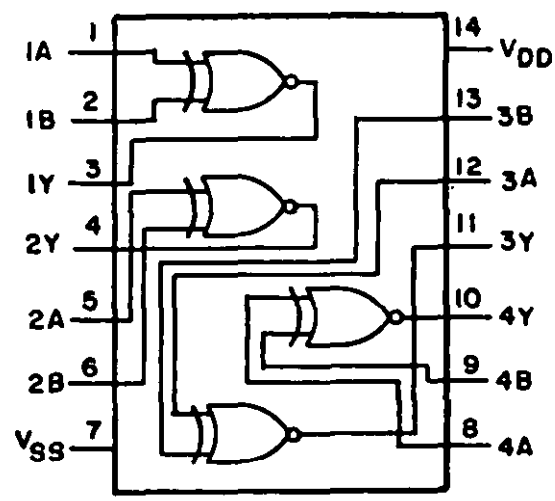


FIGURE 1. Logic diagrams and terminal connections.

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Device type 01

TRUTH TABLE					
INPUTS					OUTPUT
A	B	C	D	I	E
L	L	L	L	L	H
L	L	L	L	H	L
H	H	L	L	L	L
L	L	H	H	L	L
H	H	H	H	L	L
H	L	H	L	L	H
L	H	L	H	L	H

$E = \overline{Inhibit + AB + CD}$

Device type 02

TRUTH TABLE										
INPUTS										OUTPUT
A	B	C	D	E	F	G	H	I	E	J
L	L	L	L	L	L	L	L	L	H	H
L	L	L	L	L	L	L	L	H	H	L
L	L	L	L	L	L	L	L	L	L	L
H	H	H	H	H	H	H	H	L	H	L
H	H	L	L	L	L	L	L	L	H	L
L	L	H	H	L	L	L	L	L	H	L
L	L	H	H	L	L	L	L	L	H	L
L	L	L	L	H	H	L	L	L	H	L
L	L	L	L	L	L	H	H	L	H	L
H	L	H	L	H	L	H	L	L	H	H

$J = \overline{Inhibit + Enable + AB + CD + EF + GH}$

Device type 03

TRUTH TABLE		
INPUTS		OUTPUTS
A	B	Y
L	L	L
L	H	H
H	L	H
H	H	L

$Y = A \oplus B$

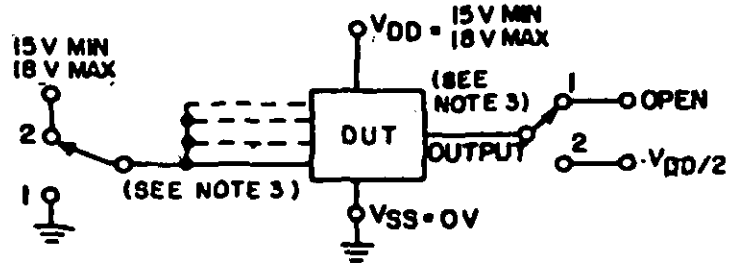
Device type 04

TRUTH TABLE		
INPUTS		OUTPUTS
A	B	$\overline{Y}$
L	L	H
L	H	L
H	L	L
H	H	H

$\overline{Y} = A \oplus B$

FIGURE 2. Truth tables and logic equations.

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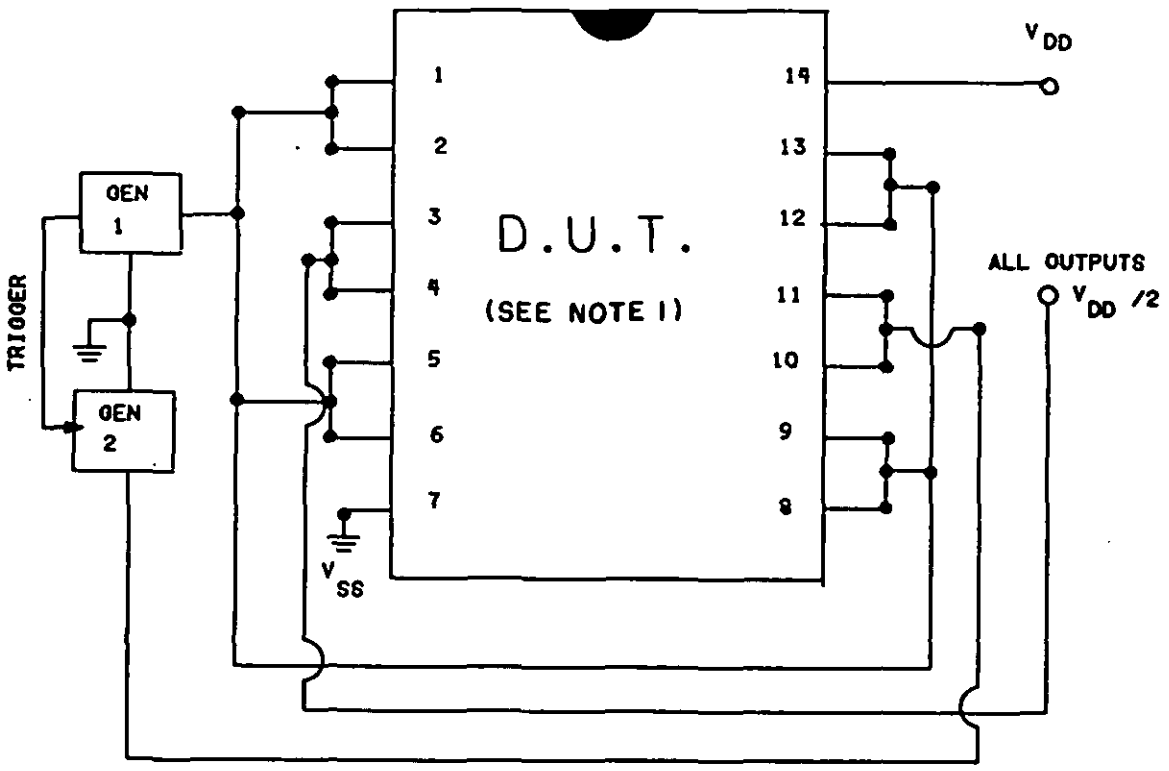


NOTES:
1. For static burn-in I, all inputs are connected to 0 volts, switch position 1.
2. For static burn-in II, all inputs are connected to V_{DD}, switch position 2.
3. Except for V_{DD} and V_{SS}, each terminal shall be connected through a resistor whose value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exceed ±20% of its branded value due to use, heat or age.
4. Output may be in switch position 1 or 2.

FIGURE 3. Static burn-in test circuits.

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Device types 01 and 02

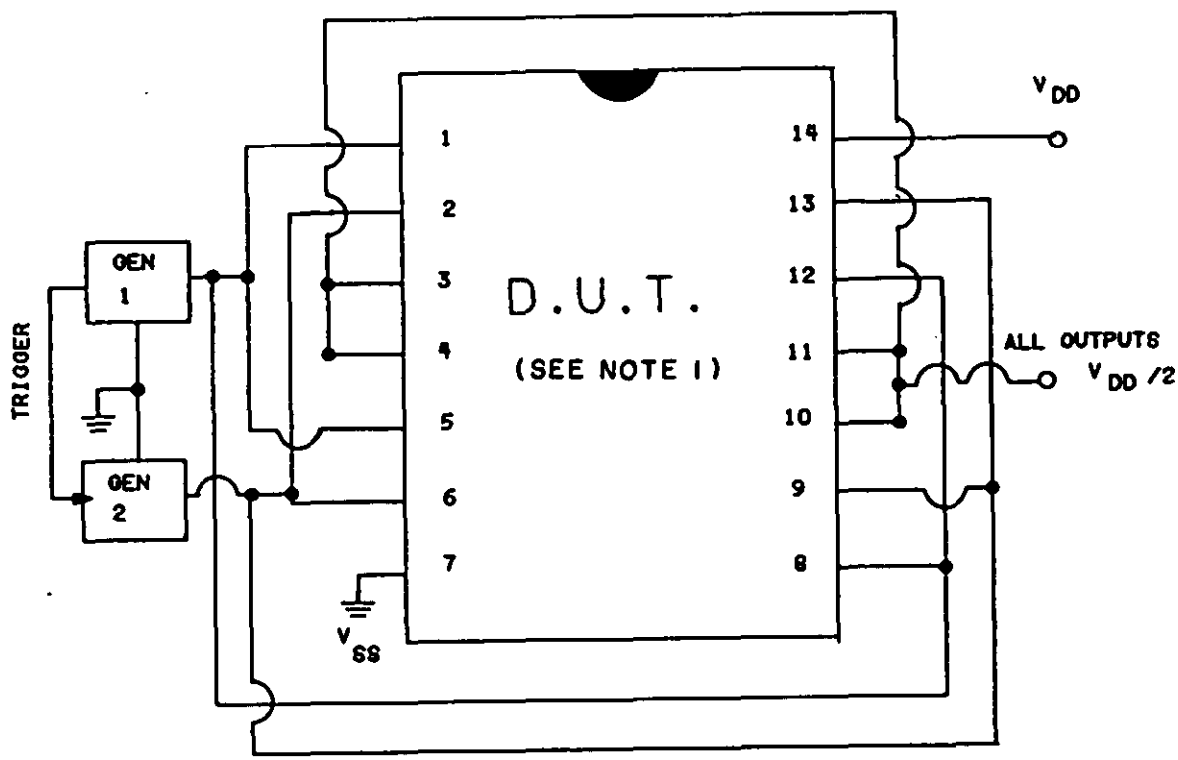


NOTES:
1. Except for V_{DD} and V_{SS}, each terminal shall be connected through a resistor whose value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exceed ±20% of its branded value due to use, heat or age.
2. Input signal requirements:
   a. Square wave, 50% duty cycle.
   b. 25 kHz ≤ PRR ≤ 1 MHz, Gen 1., Gen 2. = PRR/2.
   c. t_{TLH} and t_{THL} ≤ 1 µs.
   d. Voltage level:
      Minimum = V_{SS} -0.5 V, +10% V_{DD}.
      Maximum = V_{DD} +0.5 V, -10% V_{DD}.
3. V_{DD} = 15.0 V minimum, 18.0 V maximum.
   $V_{DD/2}$ = $V_{DD/2}$ ±1.0 V.
   V_{SS} = 0.0 V.

FIGURE 4. Dynamic burn-in and steady state life test circuit.

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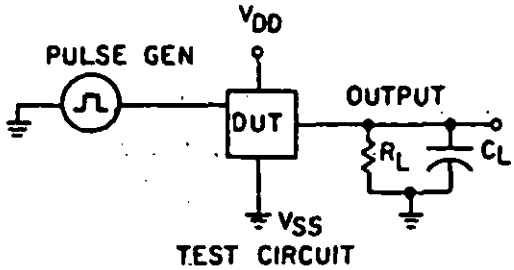
Device types 03 and 04



NOTES:
1. Except for V_{DD} and V_{SS}, each terminal shall be connected through a resistor whose value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exceed ±20% of its branded value due to use, heat or age.
2. Input signal requirements:
   a. Square wave, 50% duty cycle.
   b. 25 kHz ≤ PRR ≤ 1 MHz, Gen 1., Gen 2. = PRR/2.
   c. t_{TLH} and t_{THL} ≤ 1 µs.
   d. Voltage level:
      Minimum = V_{SS} -0.5 V, +10% V_{DD}.
      Maximum = V_{DD} +0.5 V, -10% V_{DD}.
3. V_{DD} = 15.0 V minimum, 18.0 V maximum.
   $V_{DD}/2$ = $V_{DD}/2$ ±1.0 V.
   V_{SS} = 0.0 V.

FIGURE 4. Dynamic burn-in and steady state life test circuit - Continued.

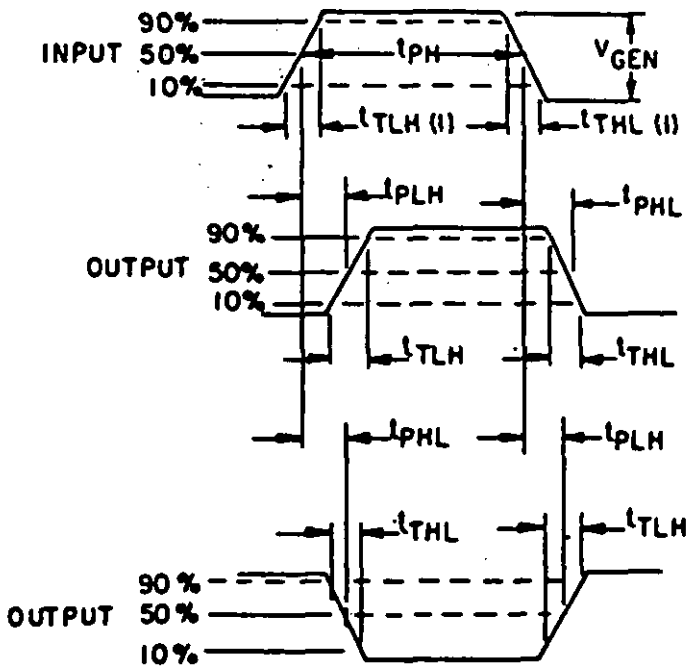
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TEST CIRCUIT

*NOTE: All unused inputs must be tied to V_{ss}.

R_L = 200 kΩ ±5%
C_L = 50 pF ±5%
(includes wiring and probe capacitance



Dynamic test waveforms

Input pulse:
$V_{GEN} = V_{DD}$ ±1.0%
t_{PH} = 1.0 ±0.1 µs
t_{THL} (1) = t_{TLH} (1) = 20 ±2 ns
PRR = 200 kHz

FIGURE 5. Switching time test circuit and waveforms.

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TABLE III. Group A inspection for device type 01.

Symbol	MIL-STD-883 method	Case A,C,B,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
		Symbol	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgp 1 T_A = 25°C		Subgp 2 T_A = 125°C		Subgp 3 T_A = -55°C		
		Test No.	1A	1B	1E	2E	2A	2B	V_{SS}	2C	2D	1I	2I	1C	1D	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{IC} (pos)		1 2 3 4 5 6 7 8 9 10	1mA	1mA			1mA	1mA		1mA	1mA	1mA	1mA	1mA	1mA	GND	1A 1B 1C 1D 1I 2I 2A 2B 2C 2D		1.5					Vdc
V_{IC} (neg)		11 12 13 14 15 16 17 18 19 20	-1mA	-1mA			-1mA	-1mA	GND	-1mA	-1mA	-1mA	-1mA	-1mA	-1mA		1A 1B 1C 1D 1I 2I 2A 2B 2C 2D		-6.0					
I_{SS} 2/	3005	21 22 23 24 25 26 27	GND GND 18.0V GND GND 18.0V 18.0V	GND GND GND 18.0V GND 18.0V 18.0V			GND GND 18.0V GND GND 18.0V 18.0V	GND GND GND 18.0V GND 18.0V 18.0V		GND GND 18.0V GND 18.0V GND 18.0V	GND GND GND 18.0V 18.0V GND 18.0V	18.0V GND ↓ 18.0V	18.0V GND ↓ 18.0V	GND GND 18.0V GND 18.0V GND 18.0V	GND GND GND 18.0V 18.0V GND 18.0V	18.0V	V_{SS}		-25		-750			nA
V_{OL1}	3007	28 29 30 31 32 33 34 35	GND 15.0V GND 15.0V GND ↓	GND 15.0V GND 15.0V GND ↓			GND ↓ 15.0V GND 15.0V	GND ↓ 15.0V GND 15.0V		GND ↓ 15.0V 15.0V	GND ↓ 15.0V 15.0V	15.0V GND ↓	GND ↓ 15.0V GND	GND GND 15.0V 15.0V GND ↓	GND GND 15.0V 15.0V GND ↓	15.0V ↓	1E ↓ 2E ↓		0.05		0.05		0.05	Vdc
V_{OH1}	3006	36 37 38 39 40 41	↓ 15.0V GND GND GND	↓ 15.0V GND GND GND			GND ↓ 15.0V GND	GND ↓ 15.0V	↓	GND ↓ 15.0V GND	GND ↓ 15.0V	↓	↓	15.0V GND GND GND	15.0V GND GND GND	↓	1E 2E 1E 1E 2E 2E	14.95 ↓		14.95 ↓		14.95 ↓		

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Case A,C,D,X,Y	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Subgp 1 T_A = 25°C Min	Max	Subgp 2 T_A = 125°C Min	Max	Subgp 3 T_A = -55°C Min	Max	Unit
		Symbol	1A	1B	1E	2E	2A	2B	V_{SS}	2C	2B	1I	2I	1C	1D	V_{DD}								
		Test No.																						
V_{IL1}		42	1.5V	1.5V			1.5V	1.5V	GND	1.5V	1.5V	1.5V	1.5V	1.5V	1.5V	5.0V	1E	4.5		4.5		4.5		Y6c
V_{IH1}		43	3.5V	1.5V										3.5V	1.5V			4.5	0.5	4.5	0.5	4.5	0.5	
		44	1.5V	3.5V										1.5V	3.5V			4.5		4.5		4.5		
		45	3.5V	3.5V										3.5V										
		46	1.5V	1.5V										3.5V										
		47	3.5V	3.5V																				
		48	3.5V	3.5V								3.5V		3.5V										
		49	1.5V	1.5V								3.5V		1.5V	1.5V									
		50										1.5V					2E	4.5		4.5		4.5		
		51					3.5V			3.5V								4.5	0.5	4.5	0.5	4.5	0.5	
		52					1.5V	3.5V		1.5V	3.5V							4.5		4.5		4.5		
		53					3.5V	1.5V		1.5V	1.5V													
		54					1.5V	3.5V		3.5V	3.5V													
		55					3.5V	3.5V		3.5V	3.5V													
		56					3.5V	3.5V		3.5V	3.5V		3.5V											
		57					1.5V	1.5V		1.5V	1.5V		3.5V											
V_{IL2}		58	3.0V	3.0V			3.0V	3.0V		3.0V	3.0V	3.0V	3.0V	3.0V	3.0V	10V	1E	9.0		9.0		9.0		
V_{IH2}		59	7.0V	3.0V										7.0V	3.0V			9.0	1.0	9.0	1.0	9.0	1.0	
		60	3.0V	7.0V										3.0V	7.0V			9.0		9.0		9.0		
		61	7.0V	7.0V										3.0V	3.0V									
		62	3.0V	3.0V										7.0V	7.0V									
		63	7.0V	7.0V										7.0V	7.0V									
		64	7.0V	7.0V								7.0V		7.0V	7.0V									
		65	3.0V	3.0V								7.0V		3.0V	3.0V									
		66								3.0V		3.0V					2E	9.0		9.0		9.0		
		67					7.0V			7.0V								9.0	1.0	9.0	1.0	9.0	1.0	
		68					3.0V	7.0V		3.0V	7.0V							9.0		9.0		9.0		
		69					7.0V			3.0V	3.0V													
		70					3.0V			7.0V	7.0V													
		71					7.0V			7.0V	7.0V													
		72					7.0V			7.0V	7.0V		7.0V											
		73					3.0V	3.0V		3.0V	3.0V		7.0V											
V_{IL3}		74	4.0V	4.0V			4.0V	4.0V		4.0V	4.0V	4.0V	4.0V	4.0V	4.0V	15.0V	1E	13.5		13.5		13.5		
V_{IH3}		75	11.0V	4.0V										11.0V	4.0V			13.5	1.5	13.5	1.5	13.5	1.5	
		76	4.0V	11.0V										4.0V	11.0V			13.5		13.5		13.5		
		77	11.0V	4.0V										4.0V	4.0V									
		78	4.0V	11.0V										11.0V	11.0V									
		79	11.0V	11.0V										11.0V	11.0V									
		80	11.0V	4.0V								11.0V		4.0V	4.0V									
		81	4.0V									11.0V					2E	13.5		13.5		13.5		
		82					11.0V			11.0V		4.0V						13.5	1.5	13.5	1.5	13.5	1.5	
		83					4.0V	11.0V		4.0V	11.0V							13.5		13.5		13.5		
		84					11.0V	11.0V		4.0V	4.0V													
		85					4.0V	4.0V		11.0V	11.0V													
		86					11.0V	11.0V		11.0V	11.0V													
		87					11.0V	11.0V		11.0V	11.0V		11.0V											
		88					11.0V	11.0V		11.0V	11.0V		11.0V											
		89					4.0V	4.0V		4.0V	4.0V													

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Case A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgp 1 T_A = 25°C		Subgp 2 T_A = 125°C		Subgp 3 T_A = -55°C		
		Symbol	1A	1B	1E	2E	2A	2B	V_{SS}	2C	2D	1I	2I	1C	1D	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																						
3/ I_{IH1}	3010	90	18.0V	18.0V			18.0V	18.0V	GND	18.0 V	18.0V	18.0V	18.0V	18.0V	18.0V	18.0V	All		10					nA
I_{IH2}		91	GND	GND			GND	GND		GND	GND	18.0V	GND	GND	GND		1I		1		45			
		92	18.0V	GND								GND		GND			1A							
		93	GND	18.0V										GND	18.0V		1B							
		94		GND										18.0V	GND		1C							
		95											18.0V	GND	18.0V		1D							
		96											GND		GND		2I							
		97					18.0V										2A							
		98					GND	18.0V									2B							
		99						GND		18.0V							2C							
		100						GND		GND	18.0V						2D							
3/ I_{IL1}	3009	101						GND		GND	GND						All		-10					
I_{IL2}		102	18.0V	18.0V			18.0V	18.0V		18.0V	18.0V		18.0V	18.0V	18.0V		1I		-1		-45			
		103	GND	18.0V								18.0V		18.0V			1A							
		104		GND										18.0V			1B							
		105		18.0V										GND			1C							
		106												18.0V			1D							
		107											GND		GND		2I							
		108					GND						18.0V		18.0V		2A							
		109					18.0V	GND									2B							
		110					18.0V	18.0V		GND							2C							
		111					18.0V	18.0V		18.0V	GND						2D							
I_{OL1}		112	GND	GND	0.4V		GND	GND		GND		5.0V	GND	GND	GND	5.0V	1E	0.51		0.36		0.64		mA
I_{OL1}		113				0.4V						GND	5.0V			5.0V	2E	0.51		0.36		0.64		
I_{OL2}		114			1.5V							15.0V	GND			15.0V	1E	3.4		2.4		4.2		
I_{OL2}		115				1.5V						GND	18.0V			15.0V	2E	3.4		2.4		4.2		
I_{OH1}		116			4.6V								GND			5.0V	1E	-0.51		-0.36		-0.64		
I_{OH1}		117				4.6V										5.0V	2E	-0.51		-0.36		-0.64		
I_{OH2}		118			13.5V											15.0V	1E	-3.4		-2.4		-4.2		
I_{OH2}		119				13.5V										15.0V	2E	-3.4		-2.4		-4.2		
																		Subgp 4 T_A = 25°C						
																		Min	Max					
CI	3012	120	4/						GND			4/				GND	1I		7.5					pF
		121		4/													1A							
		122												4/			1B							
		123											4/		4/		1C							
		124					4/										1D							
		125						4/									2I							
		126												4/			2A							
		127													4/		2B							
		128															2C							
		128A			–			—		4/	..						1D							
		128B									4/						[illegible]							
		128C															2D							

See footnotes at end of device type 01.

MIL-M-38510/128

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Case A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgp 9 T_A = 25 °C		Subgp 10 T_A = 125 °C		Subgp 11 T_A = -55 °C		
		Symbol	1A	1B	1E	2E	2A	2B	V_{SS}	2C	2D	1I	2I	1C	1D	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																						
t_{PHL}	3003 Fig. 1	130	GND	GND	OUT		GND	GND	GND	GND	GND	IN	GND	GND	GND	5.0V	1I to 1E	15	300	21	420	15	300	ns
		131	IN	5.0V								GND		GND	GND		1A to 1E	23	460	32	630	23	460	
		132	5.0V	IN										GND	GND		1B to 1E	↓	↓	↓	↓	↓	↓	
		133	GND	GND										IN	5.0V		1C to 1E							
		134												5.0V	IN		1D to 1E							
		135				OUT							IN	GND	GND		2I to 2E	15	300	21	420	15	300	
		136					IN	5.0V					GND				2A to 2E	23	460	32	630	23	460	
		137					5.0V	IN									2B to 2E							
		138					GND	GND		IN	5.0V						2C to 2E							
		139								5.0V	IN						2D to 2E							
t_{PLH}		140			OUT					GND	GND	IN					1I to 1E	25	500	35	700	25	500	
		141	IN	5.0V								GND					1A to 1E	31	620	44	870	31	620	
		142	5.0V	IN													1B to 1E	↓	↓	↓	↓	↓	↓	
		143	GND	GND										IN	5.0V		1C to 1E							
		144												5.0V	IN		1D to 1E							
		145				OUT							IN	GND	GND		2I to 2E	25	500	35	700	25	500	
		146					IN	5.0V					GND				2A to 2E	31	620	44	870	31	620	
		147					5.0V	IN									2B to 2E							
		148					GND	GND		IN	5.0V						2C to 2E							
		149								5.0V	IN						2D to 2E							
t_{THL}	3004 Fig. 5	150			OUT					GND	GND	IN					1E	10	300	14	200	10	200	
		151	IN	5.0V								GND												
		152	5.0V	IN														↓	↓	↓	↓	↓	↓	
		153	GND	GND										IN	5.0V									
		154												5.0V	IN									
		155				OUT							IN	GND	GND		2E							
		156					IN	5.0V					GND											
		157					5.0V	IN																
		158					GND	GND		IN	5.0V													
		159								5.0V	IN													
t_{TLH}		160			OUT					GND	GND	IN					1E							
		161	IN	5.0V								GND												
		162	5.0V	IN														↓	↓	↓	↓	↓	↓	
		163	GND	GND										IN	5.0V									
		164												5.0V	IN									
		165				OUT							IN	GND	GND		2E							
		166					IN	5.0V					GND											
		167					5.0V	IN																
		168					GND	GND		IN	5.0V													
		169								5.0V	IN													
																		Subgp 12 T_A = 25 °C						
t_{PHL}	3003 Fig. 5	170	GND	GND	OUT		GND	GND	GND	GND	GND	IN	GND	GND	GND	10.0V	1I to 1E	6	120					ns
		171	IN	10.0V								GND		GND	GND		1A to 1E	9	180					
		172	10.0V	IN										GND	GND		1B to 1E	↓	↓					
		173	GND	GND										IN	10.0V		1C to 1E							
		174												10.0V	IN		1D to 1E							
		175				OUT							IN	GND	GND		2I to 2E	6	120					
		176					IN	10.0V					GND				2A to 2E	9	180					
		177					10.0V	IN									2B to 2E							
		178					GND	GND		IN	10.0V						2C to 2E							
		179								10.0V	IN						2D to 2E							

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Case A,C,D,K,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
		Symbol	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgp 12 T_A = 25°C						
		Test No.	1A	1B	1E	2E	2A	2B	Vss	2C	2B	1I	2I	1C	1D	V_{DD}								
																		Min	Max	Min	Max	Min	Max	
t_{PLH}	3003 FIG. 5	180 181 182 183 184 185 186 187 188 189	GND IN 10.0 V GND	GND 10.0 V IN GND	OUT	OUT	GND IN 10.0 V GND	GND 10.0 V IN GND	GND	GND IN 10.0 V	GND 10.0 V IN	IN GND	GND IN GND	GND GND GND IN 10.0 V GND	GND GND GND GND 10.0 V IN GND	10.0V	11 to 1E 1A to 1E 1B to 1E 1C to 1E 1D to 1E 2I to 1E 2A to 2E 2B to 2E 2C to 2E 2B to 2E	10 13 10 13	200 250 200 250					ns
t_{THL}	3004 FIG. 5	190 191 192 193 194 195 196 197 198 199	IN 10.0 V GND 10.0V	10.0 V IN GND	OUT	OUT	IN 10.0 V GND	10.0 V IN GND		GND IN 10.0 V	GND 10.0 V IN	IN GND	IN GND	IN 10.0 V GND	10.0 V IN GND		1E 2E	5	100					
t_{TLH}		200 201 202 203 204 205 206 207 208 209	IN 10.0 V GND	10.0 V IN GND	OUT	OUT	IN 10.0 V GND	10.0 V IN GND		GND IN 10.0 V	GND 10.0 V IN	IN GND	IN GND	IN 10.0 V GND	10.0V IN GND		1E 2E							

1/ Pins not designated may be "HIGH" level logic, "LOW" level logic or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminals shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.

2/ The I_{SS} measurements shall be performed in sequence.

3/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.

4/ See 4.4.1c.

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TABLE III. Group A inspection for device type 02.

Symbol	MIL-STD-883 method	Case A,C,B,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgp 1 T_A = 25°C		Subgp 2 T_A = 125°C		Subgp 3 T_A = -55°C		
		Symbol	A	B	J	NC	E	F	V_{SS}	G	H	I	EN	C	D	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																						
V_{IC} (pos)		1 2 3 4 5 6 7 8 9 10	1mA	1mA			1mA	1mA		1mA	1mA	1mA	1mA	1mA	1mA	GND	A B C D E F G H I EN		1.5					Vdc
V_{IC} (neg)		11 12 13 14 15 16 17 18 19 20	-1mA	-1mA			-1mA	-1mA	GND	-1mA	-1mA	-1mA	-1mA	-1mA	-1mA		A B C D E F G H I EN		-6.0					
I_{SS} 2/	3006	21 22 23 24 25 26 27 28 29	GND GND GND 18.0V 18.0V GND GND GND 18.0V	GND GND GND 18.0V 18.0V GND			GND GND GND 18.0V GND GND 18.0V GND 18.0V	GND GND GND 18.0V GND GND 18.0V GND		GND GND GND 18.0V GND GND GND 18.0V 18.0V	GND GND GND 18.0V GND GND GND 18.0V GND	GND 18.0V GND 18.0V GND	18.0V 18.0V GND 18.0V	GND GND GND 18.0V GND GND 18.0V GND 18.0V	GND GND GND 18.0V GND 18.0V GND	18.0V	V_{SS}		-25		-750			nA
V_{OL1}	3007	30 31 32 33 34 35 36	GND GND 15.0V 15.0V GND	15.0V 15.0V GND			GND GND 15.0V GND GND 15.0V GND	15.0V GND GND 15.0V GND		GND GND 15.0V GND GND 15.0V 15.0V	GND GND 15.0V GND GND 15.0V 15.0V	15.0V GND	15.0 V GND 15.0 V	GND GND 15.0V GND 15.0V GND GND	15.0V GND 15.0V GND	15.0V	J		0.06		0.05		0.05	Vdc
V_{OH1} V_{OH1} V_{OH1}	3006 3006 3006	37 38 39	15.0V GND	15.0V			GND 15.0V GND	GND GND 15.0V		GND 15.0V GND	15.0V GND 15.0V			GND 15.0V GND	15.0V			14.95 14.95 14.95		14.95 14.95 14.95		14.95 14.95 14.95		

See footnotes at end of device type 02.

MIL-M-38510/11723

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Case A,C,D,E,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgp 1 T_A = 25°C		Subgp 2 T_A = 125°C		Subgp 3 T_A = -55°C		
		Symbol	A	B	J	NC	E	F	V_{SS}	G	H	I	EN	C	D	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																						
V_{IL1}		40	1.5V	1.5V			1.5V	1.5V	GND	1.5V	1.5V	1.5V	3.5V	1.5V	1.5V	5.0V	J	4.5		4.5		4.5		Vdc
V_{IH1}		41	1.5V	1.5V			1.5V	1.5V		1.5V	1.5V	3.5V	3.5V	1.5V	1.5V				0.5		0.5		0.5	
		42	1.5V	1.5V			3.5V	3.5V		3.5V	3.5V	3.5V	1.5V	3.5V	3.5V									
		43	3.5V	3.5V			1.5V	1.5V		1.5V	1.5V	1.5V		1.5V	1.5V									
		44	3.5V	3.5V			3.5V	3.5V		1.5V	1.5V			1.5V	1.5V									
		45	1.5V	1.5V			1.5V	1.5V		3.5V	3.5V			1.5V	1.5V									
		46	1.5V				1.5V	1.5V		1.5V	1.5V			3.5V	3.5V									
		47	3.5V				3.5V	1.5V		3.5V	1.5V			3.5V	1.5V			4.5		4.5		4.5		
		48	1.5V	3.5V			1.5V	3.5V		1.5V	3.5V			1.5V	3.5V			4.5		4.5		4.5		
V_{IL2}		50	3.0V	3.0V			3.0V	3.0V		3.0V	3.0V	3.0V	7.0V	3.0V	3.0V	10.0V		9.0		9.0		9.0		
V_{IH2}		51	3.0V	3.0V			3.0V	3.0V		3.0V	3.0V	7.0V	7.0V	3.0V	3.0V				1.0		1.0		1.0	
		52	3.0V	3.0V			3.0V	3.0V		3.0V	3.0V	3.0V	3.0V	3.0V	3.0V									
		53	7.0V	7.0V			7.0V	7.0V		7.0V	7.0V	7.0V	7.0V	3.0V	7.0V									
		54	7.0V	7.0V			3.0V	3.0V		3.0V	3.0V	3.0V		3.0V	3.0V									
		55	3.0V	3.0V			7.0V	7.0V		3.0V	3.0V			3.0V	3.0V									
		56	3.0V				3.0V	3.0V		7.0V	7.0V			3.0V	3.0V									
		57	3.0V				3.0V	3.0V		3.0V	3.0V			7.0V	7.0V									
		58	7.0V				7.0V	3.0V		7.0V	3.0V			7.0V	3.0V			9.0		9.0		9.0		
		59	3.0V	7.0V			3.0V	7.0V		3.0V	7.0V			3.0V	7.0V			9.0		9.0		9.0		
V_{IL3}		60	4.0V	4.0V			4.0V	4.0V		4.0V	4.0V	4.0V	11.0V	4.0V	4.0V	15.0V		13.5		13.5		13.5		
V_{IH3}		61	4.0V	4.0V			4.0V	4.0V		4.0V	4.0V	11.0V	11.0V	4.0V	4.0V				1.5		1.5		1.5	
		62	4.0V	4.0V			4.0V	4.0V		4.0V	4.0V	4.0V	4.0V	4.0V	4.0V									
		63	11.0V	11.0V			11.0V	11.0V		11.0V	11.0V	11.0V	11.0V	11.0V	11.0V									
		64	11.0V	11.0V			4.0V	4.0V		4.0V	4.0V	4.0V		4.0V	4.0V									
		65	4.0V	4.0V			11.0V	11.0V		4.0V	4.0V			4.0V	4.0V									
		66	4.0V				4.0V	4.0V		11.0V	11.0V			4.0V	4.0V									
		67	4.0V				4.0V	4.0V		4.0V	4.0V			11.0V	11.0V									
		68	11.0V				11.0V	4.0V		11.0V	4.0V			11.0V	4.0V			13.5		13.5		13.5		
		69	4.0V	11.0V			4.0V	11.0V		4.0V	11.0V			4.0V	11.0V			13.5		13.5		13.5		
I_{IH1} 2/	3010	70	18.0V	18.0V			18.0V	18.0V		18.0V	18.0V	18.0V	18.0V	18.0V	18.0V	18.0V	All inputs together		10					nA
I_{IH2}		71	18.0V	GND			GND	GND		GND	GND	GND	GND	GND	GND		A		1		45			
		72	GND	18.0V										GND	GND		B							
		73		GND										GND	GND		C							
		74												18.0V	18.0V		D							
		75					18.0V							GND	GND		E							
		76					GND	18.0V									F							
		77						GND		18.0V	18.0V						G							
		78								GND	GND	18.0V					H							
		79								GND	GND	GND	18.0V				I							
		80															EN							

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Case A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	A	B	J	NC	E	F	V_{SS}	G	H	I	EN	C	D	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																						
I_{IL1} 2/	3009	81	GND	GND			GND	GND	GND	GND	GND	GND	GND	GND	GND	18.0V	All inputs together		-10					nA
I_{IL2}		82 83 84 85 86 87 88 89 90 91	GND 18.0V	18.0V GND 18.0V			18.0V GND 18.0V	18.0V GND 18.0V		18.0V GND 18.0V 18.0V 18.0V	18.0V GND 18.0V 18.0V	18.0V GND 18.0V	18.0V GND	18.0V 18.0V GND 18.0V	18.0V 18.0V GND 18.0V		A B C D E F G H I EN		-1		-20			
I_{OL1}		92	GND	GND	0.4V		GND	GND		GND	GND	GND	GND	GND	GND	5.0V	J	0.51		0.36		0.64		mA
I_{OL2}		93			1.5V								GND			15.0V		3.4		2.4		4.2		
I_{OH1}		94			4.6V								5.0V			5.0V		-0.51		-0.36		-0.64		
I_{OH2}		95			13.5V								15.0V			15.0V		-3.4		-2.4		-4.2		
																		Subgroup 4 T_A = 25°C						
CI	3012	96 97 98 99 100 101 102 103 104 105	4/	4/			4/	4/	GND	4/	4/	4/	4/	4/	4/	GND	A B C D E F G H I EN		7.5					pF
																		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
t_{PHL}	3003 Fig. 5	106 107 108 109 110 111 112 113 114 115	GND IN 5.0V GND	GND 5.0V IN GND	OUT		GND IN 5.0V GND	GND 5.0V IN GND	GND	GND IN 5.0V GND	GND 5.0V IN GND	IN GND	5.0V IN	GND GND GND IN 5.0V GND	GND GND GND 5.0V IN GND	5.0V	I to J A to J B to J C to J D to J E to J F to J G to J H to J EN to J	15 23	300 450	21 32	420 630	15 23	300 450	ns
t_{PLH}		116 117 118 119 120 121 122 123 124 125	IN 5.0V GND	5.0V IN GND			IN 5.0V GND GND GND	5.0V IN GND GND GND		IN 5.0V GND	5.0V IN GND	IN GND	5.0V IN	IN 5.0V GND	5.0V IN GND		I to J A to J B to J C to J D to J E to J F to J G to J H to J EN to J	25 35	500 700	35 49	700 980	25 35	500 700	

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Case A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgp 9 T_A=25°C		Subgp 10 T_A=125°C		Subgp 11 T_A=-55°C		
		Symbol	A	B	J	NC	E	F	V_{SS}	G	H	I	EN	C	D	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																						
t_{THL}	3004 Fig. 5	126	GND	GND	OUT		GND	GND	GND	GND	GND	IN	5.0V	GND	GND	5.0V	J	10	100	15	150	10	100	ns
		127	IN	5.0V								GND		GND	GND									
		128	5.0V	IN										GND	GND									
		129	GND	GND										IN	5.0V									
		130												5.0V	IN									
		131					IN	5.0V						GND	GND									
		132					5.0V	IN																
		133					GND	GND		IN	5.0V													
		134								5.0V	IN													
		135								GND	GND		IN											
t_{TLH}		136										IN	5.0V											
		137	IN	5.0V								GND		GND	GND									
		138	5.0V	IN										IN	5.0V									
		139	GND	GND										5.0V	IN									
		140												GND	GND									
		141					IN	5.0V																
		142					5.0V	IN																
		143					GND	GND		IN	5.0V													
		144					GND	GND		5.0V	IN													
		145					GND	GND		GND	GND		IN											
																		Subgp 12 T_A=25°C						
t_{PHL}	3003 Fig. 5	146	GND	GND	OUT		GND	GND	GND	GND	GND	IN	10.0V	GND	GND	10.0V	I to J	0	110					ns
		147	IN	10.0V								GND		GND	GND		A to J	9	150					
		148	10.0V	IN										GND	GND		B to J							
		149	GND	GND										IN	10.0V		C to J							
		150												10.0V	IN		D to J							
		151					IN	10.0V						GND	GND		E to J							
		152					10.0V	IN									F to J							
		153					GND	GND		IN	10.0V						G to J							
		154								10.0V	IN						H to J							
		155								GND	GND		IN				EN to J							
t_{PLH}		156										IN	10.0V				I to J	10	200					
		157	IN	10.0V								GND		GND	GND		A to J	14	250					
		158	10.0V	IN										GND	GND		B to J							
		159	GND	GND										IN	10.0V		C to J							
		160												10.0V	IN		D to J							
		161					IN	10.0V						GND	GND		E to J							
		162					10.0V	IN									F to J							
		163					GND	GND		IN	10.0V						G to J							
		164								10.0V	IN						H to J							
		165								GND	GND		IN				EN to J							
t_{THL}	3004 Fig. 5	166										IN	10.0V				J	5	100					
		167	IN	10.0V								GND		GND	GND									
		168	10.0V	IN										GND	GND									
		169	GND	GND										IN	10.0V									
		170												10.0V	IN									
		171					IN	10.0V						GND	GND									
		172					10.0V	IN																
		173					GND	GND		IN	10.0V													
		174					GND	GND		10.0V	IN													
		175					GND	GND		GND	GND		IN											

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Case A,C,D,E,Y	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Subgp 12 T_A = 25°C Min	Max	Min	Max	Min	Max	Unit
		Symbol	A	B	J	NC	E	F	V_{SS}	G	H	I	EN	C	D	V_{DD}								
		Test No.																						
t_{TLH}	3004 Fig. 5	176	GND	GND	OUT		GND	GND	GND	GND	GND	IN	10.0V	GND	GND	10.0V	J	5	100					ns
		177	IN	10.0V IN								GND		GND	GND									
		178	10.0V	GND										GND	GND									
		179	GND											IN	10.0V									
		180					IN	10.0V						10.0V	IN									
		181					10.0V	IN		IN	10.0V		10.0V	GND	GND									
		182					GND	GND		10.0V	IN		IN											
		183					GND	GND		GND	GND													
		184					GND	GND																
		185					GND	GND																

1/ Pins not designated may be "HIGH" level logic, "Low" level logic or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminals shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.

2/ The I_{SS} measurements shall be performed in sequence.

3/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.

4/ See 4.4.1c.

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TABLE III. Group A inspection for device type 02.

Symbol	MIL-STD-883 method	Case A,C,D,E,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgp 1 T_A = 25°C		Subgp 2 T_A = 125°C		Subgp 3 T_A = -55°C		
		Symbol	1A	1B	1Y	2Y	2A	2B	V_{SS}	3A	3B	3Y	4Y	4A	4B	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																						
V_{IC} (pos)		1 2 3 4 5 6 7 8	1mA	1mA			1mA	1mA		1mA	1mA			1mA	1mA	GND	1A 1B 2A 2B 3A 3B 4Y 4B		1.5					Vdc
V_{IC} (neg)		9 10 11 12 13 14 15 16	-1mA	-1mA			-1mA	-1mA	GND	-1mA	-1mA			-1mA	-1mA		1A 1B 2A 2B 3A 3B 4A 4B		-6.0					
I_{SS} 2/	3005	17 18 19 20	GND GND 18.0V 18.0V	GND 18.0V GND 18.0V			GND GND 18.0V 18.0V	GND 18.0V GND 18.0V		GND GND 18.0V 18.0V	GND 18.0V GND 18.0V			GND GND 18.0V 18.0V	GND 18.0V GND 18.0V	18.0V	V_{SS}		-25		-750			nA
V_{OL1}	3007	21 22 23 24 25 26 27 28	GND 15.0V	GND 15.0V			GND 15.0V	GND 15.0V		GND 15.0V	GND 15.0V			GND 15.0V	GND 15.0V	15.0V	1Y 2Y 3Y 4Y 1Y 2Y 3Y 4Y		0.05		0.05		0.05	Vdc
V_{OH1}	3006	29 30 31 32 33 34 35 36	GND	GND 15.0V GND			GND GND 15.0V GND	GND GND GND 15.0V GND		GND 15.0V GND	GND 15.0V GND			GND 15.0V GND	GND 15.0V		1Y 1Y 2Y 2Y 3Y 3Y 4Y 4Y	14.95		14.95		14.95		
V_{IH1}		37 38 39 40 41 42 43 44	3.5V 1.5V GND	1.5V 3.5V GND			3.5V 1.5V GND	1.5V 3.5V GND		3.5V 1.5V GND GND	1.5V 3.5V GND GND			3.5V 1.5V	GND 1.5V 3.5V	5.0V	1Y 1Y 2Y 2Y 3Y 3Y 4Y 4Y	4.5		4.5		4.5		

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Case A, C, D, X, Y	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Subgp 1 T_A = 25°C Min	Max	Subgp 2 T_A = 125°C Min	Max	Subgp 3 T_A = -55°C Min	Max	Unit
		Symbol	1A	1B	1Y	2Y	2A	2B	V_{SS}	3A	3B	3Y	4Y	4A	4B	V_{DD}								
		Test No.																						
V_{IH2}		45 46 47 48 49 50 51 52	7.0V 3.0V GND	3.0V 7.0V GND			GND GND 7.0V 3.0V GND	GND GND 3.0V 7.0V GND	GND	GND 7.0V 3.0V GND	GND 3.0V 7.0V GND			GND 7.0V 3.0V	GND 3.0V 7.0V	10.0V	1Y 1Y 2Y 2Y 3Y 3Y 4Y 4Y	9.0V		9.0V		9.0V		Vdc
V_{IH3}		53 54 55 56 57 58 59 60	11.0V 4.0V GND	4.0V 11.0V GND			11.0V 4.0V GND	4.0V 11.0V GND		11.0V 4.0V GND	4.0V 11.0V GND			GND 11.0V 4.0V	GND 4.0V 11.0V	15.0V	1Y 1Y 2Y 2Y 3Y 3Y 4Y 4Y	13.5		13.5		13.5		
V_{IL1}		61 62 63 64	1.5V GND GND GND				1.5V GND GND			1.5V GND				GND GND GND 1.5V	GND	5.0V	1Y 2Y 3Y 4Y		0.5		0.5		0.5	
V_{IL2}		65 66 67 68	3.0V GND GND GND				GND 3.0V GND GND			GND GND 3.0V GND				GND GND GND 3.0V		10.0V	1Y 2Y 3Y 4Y		1.0		1.0		1.0	
V_{IL3}		69 70 71 72	4.0V GND GND GND				GND 4.0V GND GND			GND GND 4.0V GND				GND GND GND 4.0V		15.0V	1Y 2Y 3Y 4Y		1.5		1.5		1.5	
I_{IH1} 3/	3010	73	18.0V	18.0V			18.0V	18.0V		18.0V	18.0V			18.0V	18.0V	18.0V	All inputs together		6					nA
I_{IH2}		74 75 76 77 78 79 80 81	18.0V GND	GND 18.0V GND			GND GND 18.0V GND	GND GND GND 18.0V GND		GND 18.0V GND	GND 18.0V GND GND			GND 18.0V GND	GND 18.0V		1A 1B 2A 2B 3A 3B 4A 4B		1		45			
I_{IL1} 3/	3009	82									GND			GND	GND		All inputs together		-6					
I_{IL2}		83 84 85 86 87 88 89 90	18.0V	18.0V GND 18.0V			18.0V 18.0V GND 18.0V	18.0V 18.0V 18.0V GND 18.0V		18.0V GND 18.0V 18.0V 18.0V	18.0V GND 18.0V 18.0V 18.0V			18.0V GND 18.0V	18.0V GND		1A 1B 2A 2B 3A 3B 4A 4B		-1		-45			

See footnotes at end of device type 03.

MIL-M-38510/128

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TABLE III. Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 method	Case A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test limits						Units
		Symbol	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgp 1 T_A=25°C		Subgp 2 T_A=125°C		Subgp 3 T_A=-55°C		
		Test No.	1A	1B	1Y	2Y	2A	2B	V_{SS}	3A	3B	3Y	4Y	4A	4B	V_{DD}		Min	Max	Min	Max	Min	Max	
I_{OL1}		91 92 93 94	GND	GND	0.4V	0.4V	GND	GND	GND	GND	GND	0.4V	0.4V	GND	GND	5.0V	1Y 2Y 3Y 4Y	0.51		0.36		0.64		mA
I_{OH1}		95 96 97 98	5.0V		4.6V	4.6V	5.0V			5.0V		4.6V	4.6V	5.0V			1Y 2Y 3Y 4Y	-0.51		-0.36		-0.64		
I_{OL2}		99 100 101 102	GND		1.5V	1.5V	GND			GND		1.5V	1.5V	GND		15.0V	1Y 2Y 3Y 4Y	3.4		2.4		4.2		
I_{OH2}		103 104 105 106	15.0V		13.5V	13.5V	15.0V			15.0V		13.5V	13.5V	13.5V			1Y 2Y 3Y 4Y	-3.4		-2.4		-4.2		
																		Subgp 4 T_A=25°C						
																		Min	Max					
C_I	3012	107 108 109 110 111 112 113 114	4/	4/			4/	4/	GND	4/	4/			4/	4/	GND	1A 1B 2A 2B 3A 3B 4A 4B		7.5					pF
																		Subgp 9 T_A=25°C		Subgp 10 T_A=125°C		Subgp 11 T_A=-55°C		
																		Min	Max	Min	Max	Min	Max	
t_{PHL}	3003 Fig. 5	115 116 117 118 119 120 121 122	IN GND	GND IN GND	OUT OUT	OUT OUT	GND GND IN GND	GND GND GND IN GND	GND	GND IN GND	GND IN GND	OUT OUT	OUT OUT	GND IN GND	GND IN	5.0V	1A to 1Y 1B to 1Y 2A to 2Y 2B to 2Y 3A to 3Y 3B to 3Y 4A to 4Y 4B to 4Y	13	250	18	360	13	250	ns
t_{PLH}	3003 Fig. 5	123 124 125 126 127 128 129 130	IN GND	IN GND	OUT OUT	OUT OUT	IN GND	IN GND		IN GND GND GND	IN GND GND GND	OUT OUT	OUT OUT	IN GND	GND IN		1A to 1Y 1B to 1Y 2A to 2Y 2B to 2Y 3A to 3Y 3B to 3Y 4A to 4Y 4B to 4Y							

See footnotes at end of device type 03.

MIL-M-38510/1128

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TABLE III. Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 method	Case A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgp 9 T_A=25°C		Subgp 10 T_A=125°C		Subgp 11 T_A=-55°C		
		Symbol	1A	1B	1Y	2Y	2A	2B	V_{SS}	3A	3B	3Y	4Y	4A	4B	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																						
t_{THL}	3004	131	IN	GND	OUT		GND	GND	GND	GND	GND			GND	GND	5.0V	1Y	10	200	18	280	10	200	ns
		132	GND	IN	OUT		GND	GND									1Y							
		133		GND		OUT	IN	GND									2Y							
		134				OUT	GND	IN									2Y							
		135						GND		IN		OUT					3Y							
		136								GND	IN	OUT					3Y							
		137									GND		OUT	IN			4Y							
		138											OUT	GND	IN		4Y							
t_{TLH}		139	IN		OUT										GND		1Y							
		140	GND	IN	OUT		IN										1Y							
		141		GND		OUT	GND	IN									2Y							
		142				OUT		GND		IN		OUT					2Y							
		143								GND	IN	OUT					3Y							
		144									GND		OUT	IN			3Y							
		145									GND		OUT	GND	IN		4Y							
		146									GND		OUT	GND	IN		4Y							
																		Subgp 12 T_A=25°C						
																		Min	Max					
t_{PHL}	3003 Fig. 5	147	IN	GND	OUT		GND	GND	GND	GND	GND			GND	GND	10.0V	1A to 1Y	6	120					ns
		148	GND	IN	OUT		GND	GND									1B to 1Y							
		149		GND		OUT	IN	GND									2A to 2Y							
		150				OUT	GND	IN									2B to 2Y							
		151						GND		IN		OUT					3A to 3Y							
		152								GND	IN	OUT					3B to 3Y							
		153									GND		OUT	IN			4A to 4Y							
		154											OUT	GND	IN		4B to 4Y							
t_{PLH}		155	IN		OUT										GND		1A to 1Y							
		156	GND	IN	OUT		IN										1B to 1Y							
		157		GND		OUT	GND	IN									2A to 2Y							
		158				OUT		GND		IN		OUT					2B to 2Y							
		159								GND	IN	OUT					3A to 3Y							
		160									GND		OUT	IN			3B to 3Y							
		161											OUT	GND	IN		4A to 4Y							
		162											OUT	GND	IN		4B to 4Y							
t_{THL}	3004 Fig. 5	163	IN		OUT									GND	GND		1Y	5	100					
		164	GND	IN	OUT		IN										1Y							
		165		GND		OUT	GND	IN									2Y							
		166				OUT		GND		IN		OUT					2Y							
		167								IN		OUT					3Y							
		168								GND	IN	OUT					3Y							
		169								GND	GND		OUT	IN			4Y							
		170								GND	GND		OUT	GND	IN		4Y							

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TABLE III. Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 method	Case A,C,D,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgp 12 T_A = 25°C						
		Symbol	1A	1B	1Y	2Y	2A	2B	V_{SS}	3A	3B	3Y	4Y	4A	4B	V_{DD}								
		Test No.																Min	Max	Min	Max	Min	Max	
t_{TLH}	3004 FIG. 5	171 172 173 174 175 176 177 178	IN GND	GND IN GND	OUT OUT	OUT OUT	GND GND IN GND	GND GND GND IN GND	GND	GND IN GND GND GND	GND IN GND GND	OUT OUT	OUT OUT	GND IN GND	GND IN	10.0V	1Y 1Y 2Y 2Y 3Y 3Y 4Y 4Y	5	100			-		ns

1/ Pins not designated may be "HIGH" level logic, "LOW" level logic or OPEN. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminals shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.

2/ The I_{SS} measurements shall be performed in sequence.

3/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.

4/ See 4.4.1c.

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TABLE III. Group A inspection for device type 06.

Symbol	MIL-STD-883 method	Case A,C,D,X,Y	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Subgp 1 T_A = 25°C Min	Max	Subgp 2 T_A = 125°C Min	Max	Subgp 3 T_A = -55°C Min	Max	Unit
		Symbol	1A	1B	1T	1Y	2A	2B	V_{SS}	4A	4B	4T	3T	3A	3B	V_{DD}								
		Test No.																						
V_{IC} (pos)		1 2 3 4 5 6 7 8	1mA	1mA			1mA	1mA		1mA	1mA			1mA	1mA	GND	1A 2A 3A 4A 1B 2B 3B 4B		1.5					Vdc
V_{IC} (neg)		9 10 11 12 13 14 15 16	-1mA	-1mA			-1mA	-1mA	GND	-1mA	-1mA			-1mA	-1mA		1A 2A 3A 4A 1B 2B 3B 4B		-6.0					
I_{SS} 2/	3006	17 18 19 20	GND GND 18.0V 18.0V	GND 18.0V GND 18.0V			GND GND 18.0V 18.0V	GND 18.0V GND 18.0V		GND GND 18.0V 18.0V	GND 18.0V GND 18.0V			GND GND 18.0V 18.0V	GND 18.0V GND 18.0V	18.0V	V_{SS}		-25		-750			nA
V_{OH1}	3006	21 22 23 24 25 26 27 28	GND 15.0V	GND 15.0V			GND 15.0V	GND 15.0V		GND 15.0V	GND 15.0V			GND 15.0V	GND 15.0V	15.0V	1Y 2Y 3Y 4Y 1T 2T 3T 4T	14.95		14.95		14.95		Vdc
V_{OL1}	3007	29 30 31 32 33 34 35 36	GND	GND 15.0V GND			GND GND 15.0V GND	GND GND GND 15.0V GND		GND 15.0V GND	GND 15.0V GND			GND 15.0V GND	GND 15.0V		1Y 2Y 3Y 4Y 1T 2T 3T 4T		0.05		0.05		0.05	
V_{IH1}		37 38 39 40 41 42 43 44	3.5V 1.5V GND	1.5V 3.5V GND			3.5V 1.5V GND	1.5V 3.5V GND		3.5V 1.5V GND GND	1.5V 3.5V GND GND			3.5V 1.5V	GND 1.5V 3.5V	5.0V	1Y 2Y 3Y 4Y 1T 2T 3T 4T		0.5		0.5		0.5	

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.

Symbol	MIL-STD-883 method	Case A,C,D,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgp 1 T_A = 25°C		Subgp 2 T_A = 125°C		Subgp 3 T_A = -55°C		
		Symbol	1A	1B	1Y	2Y	2A	2B	V_{SS}	4A	4B	4Y	3Y	3A	3B	V_{DD}								
		Test No.																Min	Max	Min	Max	Min	Max	
V_{IH2}		45 46 47 48 49 50 51 52	7.0V 3.0V GND	3.0V 7.0V GND			GND GND 7.0V 3.0V GND	GND GND 3.0V 7.0V GND	GND	GND 7.0V 3.0V GND	GND 3.0V 7.0V GND			GND 7.0V 3.0V	GND 3.0V 7.0V	10.0V	1Y 1Y 2Y 2Y 4Y 4Y 3Y 3Y		1.0		1.0		1.0	Vdc
V_{IH3}		53 54 55 56 57 58 59 60	11.0V 4.0V GND	4.0V 11.0V GND			11.0V 4.0V GND	4.0V 11.0V GND		11.0V 4.0V GND	4.0V 11.0V GND			GND 11.0V 4.0V	GND 4.0V 11.0V	15.0V	1Y 1Y 2Y 2Y 4Y 4Y 3Y 3Y		1.5		1.5		1.5	
V_{IL1}		61 62 63 64	1.5V GND GND GND				1.5V GND GND			1.5V GND				GND GND GND 1.5V	GND	5.0V	1Y 2Y 4Y 3Y	4.5		4.5		4.5		
V_{IL2}		65 66 67 68	3.0V GND GND GND				GND 3.0V GND GND			GND 3.0V GND				GND GND GND 3.0V		10.0V	1Y 2Y 4Y 3Y	9.0		9.0		9.0		
V_{IL3}		69 70 71 72	4.0V GND GND GND				GND 4.0V GND GND			GND GND 4.0V GND				GND GND GND 4.0V		15.0V	1Y 2Y 4Y 3Y	13.5		13.5		13.5		
I_{IH1} 3/	3010	73	18.0V	18.0V			18.0V	18.0V		18.0V	18.0V			18.0V	18.0V	18.0V	All inputs together		8					nA
I_{IH2}		74 75 76 77 78 79 80 81	18.0V GND	GND 18.0V GND			GND GND 18.0V GND	GND GND GND 18.0V GND		GND 18.0V GND	GND 18.0V GND GND			GND 18.0V GND	GND 18.0V		1A 1B 2A 2B 4A 4B 3A 3B		1		45			
I_{IL1} 3/	3009	82									GND			GND	GND	GND	All inputs together		-8					

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 -Continued.

Symbol	MIL-STD-883 method	Case A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgp 1 T_A = 25°C		Subgp 2 T_A=125°C		Subgp 3 T_A=-55°C		
		Symbol	1A	1B	1Y	2Y	2A	2B	V_{SS}	4A	4B	4Y	3Y	3A	3B	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																						
I_{IL2}	3009	83 84 86 86 87 88 89 90	GND 18.0V	18.0V GND 18.0V			18.0V 18.0V GND 18.0V	18.0V 18.0V 18.0V GND 18.0V 18.0V 18.0V GND	GND	18.0V GND 18.0V 18.0V 18.0V	18.0V GND 18.0V 18.0V			18.0V GND 18.0V	18.0V GND	18.0V	1A 1B 2A 2B 4A 4B 3A 3B		-1		-45			nA
I_{OH1}		91 92 93 94	GND	GND	4.6V	4.6V	GND			GND	GND	4.6V	4.6V	GND		5.0V	1Y 2Y 4Y 3Y	-0.51		-0.36		-0.64		mA
I_{OL1}		95 96 97 98	5.0V		0.4V	0.4V	5.0V			5.0V		0.4 V	0.4V	5.0V			1Y 2Y 4Y 3Y	0.51		0.36		0.64		
I_{OH2}		99 100 101 102	GND		13.5V	13.5V	GND			GND		13.5V	13.5V	GND		15.0V	1Y 2Y 4Y 3Y	-3.4		-2.4		-4.2		
I_{OL2}		103 104 105 106	15.0V		1.5V	1.5V	15.0V			15.0V		1.5V	1.5V	15.0V			1Y 2Y 4Y 3Y	3.4		2.4		4.2		
																		Subgp 4 T_A=25°C						
																		Min	Max					
C_I	3012	107 108 109 110 111 112 113 114	4/	4/			4/	4/	GND	4/	4/			4/	4/	GND	1A 1B 2A 2B 4A 4B 3A 3B		7.5					pF
																		Subgp 9 T_A=25°C		Subgp 10 T_A=125°C		Subgp 11 T_A=-55°C		
																		Min	Max	Min	Max	Min	Max	
t_{PHL}	3003 Fig. 5	115 116 117 118 119 120 121 122	IN GND	GND IN GND	OUT OUT	OUT OUT	GND GND IN GND	GND GND GND IN GND	GND	GND IN GND GND GND	GND IN GND GND	OUT OUT	OUT OUT	GND IN GND	GND IN	5.0V	1A to 1Y 1B to 1Y 2A to 2Y 2B to 2Y 4A to 4Y 4B to 4Y 3A to 3Y 3B to 3Y	13	250	18	250	13	250	ns

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TABLE III. Group A inspection for device type 04 -Continued.

Symbol	MIL-STD-883 method	Case A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgp 9 T_A=25°C		Subgp 10 T_A=125°C		Subgp 11 T_A=-55°C		
		Symbol	1A	1B	1Y	2Y	2A	2B	V_{SS}	4A	4B	4Y	3Y	3A	3B	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																						
t_{PLH}	3003 Fig. 5	123	IN	GND	OUT		GND	GND	GND	GND	GND			GND	GND	5.0V	1A to 1Y	13	250	18	350	13	250	ns
		124	GND	IN	OUT		GND	GND									1B to 1Y							
		125		GND		OUT	IN	GND									2A to 2Y							
		126				OUT	GND	IN									2B to 2Y							
		127						GND		IN		OUT					4A to 4Y							
		128								GND	IN	OUT					4B to 4Y							
		129									GND		OUT	IN			3A to 3Y							
		130											OUT	GND	IN		3B to 3Y							
t_{THL}	3004 Fig. 5	131	IN		OUT										GND		1Y	10	300	14	280	10	200	
		132	GND	IN	OUT												1Y							
		133		GND		OUT	IN										2Y							
		134				OUT	GND	IN									2Y							
		135						GND		IN		OUT					4Y							
		136								GND	IN	OUT					4Y							
		137									GND		OUT	IN			3Y							
		138											OUT	GND	IN		3Y							
t_{TLH}		139	IN		OUT										GND		1Y							
		140	GND	IN	OUT												1Y							
		141		GND		OUT	IN										2Y							
		142				OUT	GND	IN									2Y							
		143						GND		IN		OUT					4Y							
		144								GND	IN	OUT					4Y							
		145								GND	GND		OUT	IN			3Y							
		146								GND	GND		OUT	GND	IN		3Y							
																		Subgp 12 T_A=25°C						
																		Min	Max					
t_{PHL}	3003 Fig. 5	147	IN	GND	OUT		GND	GND	GND	GND	GND			GND	GND	10.0V	1A to 1Y	8	120					ns
		148	GND	IN	OUT		GND	GND									1B to 1Y							
		149		GND		OUT	IN	GND									2A to 2Y							
		150				OUT	GND	IN									2B to 2Y							
		151						GND		IN		OUT					4A to 4Y							
		152								GND	IN	OUT					4B to 4Y							
		153									GND		OUT	IN			3A to 3Y							
		154											OUT	GND	IN		3B to 3Y							
t_{PLH}		155	IN		OUT										GND		1A to 1Y							
		156	GND	IN	OUT												1B to 1Y							
		157		GND		OUT	IN										2A to 2Y							
		158				OUT	GND	IN									2B to 2Y							
		159						GND		IN		OUT					4A to 4Y							
		160								GND	IN	OUT					4B to 4Y							
		161								GND	GND		OUT	IN			3A to 3Y							
		162								GND	GND		OUT	GND	IN		3B to 3Y							

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TABLE III. Group A inspection for device type 04 - Continued.

Symbol	MIL-STD-883 method	Case A,C,D,K,Y	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Subgp 12 T_A = 25°C Min	Max	Min	Max	Min	Max	Unit
		Symbol	1A	1B	1Y	2Y	2A	2B	V_{SS}	4A	4B	4Y	3Y	3A	3B	V_{DD}								
		Test No.																						
t_{THL}	3004 FIG. 5	163	IN	GND	OUT	OUT	GND	GND	GND	GND	GND	OUT	OUT	GND	GND	10.0V	1Y	5	100					ns
		164	GND	IN	OUT	OUT	GND	GND		GND	GND	OUT	OUT	GND	GND		1Y							
		165	GND	GND	OUT	OUT	IN	GND		IN	GND	OUT	OUT	GND	GND		2Y							
		166					GND	GND		GND	GND	OUT	OUT	GND	GND		2Y							
		167					GND	GND		GND	GND	OUT	OUT	GND	GND		4Y							
		168					GND	IN		GND	GND	OUT	OUT	GND	GND		4Y							
		169					GND	GND		GND	GND	OUT	OUT	GND	GND		3Y							
		170					GND	GND		GND	GND	OUT	OUT	GND	GND		3Y							
t_{TLH}		171	IN	GND	OUT	OUT	GND	GND		GND	GND	OUT	OUT	GND	GND		1Y							
		172	GND	IN	OUT	OUT	GND	GND		GND	GND	OUT	OUT	GND	GND		1Y							
		173	GND	GND	OUT	OUT	IN	GND		IN	GND	OUT	OUT	GND	GND		2Y							
		174					GND	GND		GND	GND	OUT	OUT	GND	GND		2Y							
		175					GND	IN		GND	GND	OUT	OUT	GND	GND		4Y							
		176					GND	GND		GND	GND	OUT	OUT	GND	GND		4Y							
		177					GND	GND		GND	GND	OUT	OUT	GND	GND		3Y							
		178					GND	GND		GND	GND	OUT	OUT	GND	GND		3Y							

1/ Pins not designated may be "HIGH" level logic, "Low" level logic or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminals shall be open; $V_{IC\ (neg)}$ tests, the V_{DD} terminals shall be open; I_{SS} tests, the output terminals shall be open.

2/ The I_{SS} measurements shall be performed in sequence.

3/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.

4/ See 4.4.1c.

MIL-M-38510/1228

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c. End-point electrical parameters shall be as specified in table II herein and shall consist only of those subgroups specified in table IIa of test method 5005 of MIL-STD-883, and table II herein also. Delta limits shall apply only to subgroup 5 of group B inspections and shall consist of tests specified in table IV herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection, and shall consist of tests specified in table IV herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D and as specified in 4.5.2 and as shown on figure 6, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

c. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.4.5 Group E inspection. Group E inspection is required only for device types intended to be marked as radiation hardened (see 3.6.1). When group E testing is performed it shall be in accordance with table V of method 5005 of MIL-STD-883 and 4.5.5 herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit V_{SS} terminal, unless otherwise specified. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cooldown procedures. When these tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to an ambient temperature of 25°C ±3°C; then, electrical parameter end-point measurements shall be performed.

TABLE IV. Delta limits at 25°C.

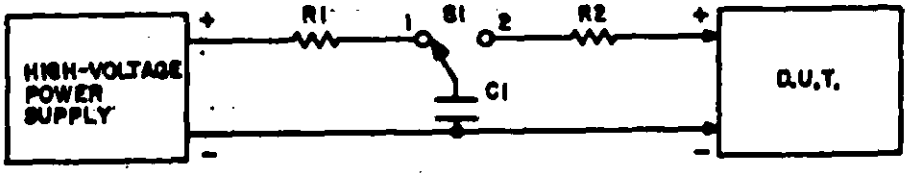
Parameter 1/	Device type			
	01	02	03	04
I_{SS}	±10 nA	±10 nA	±10 nA	±10 nA
I_{OL1}	±15%	±15%	±15%	±15%
I_{OH1}	±15%	±15%	±15%	±15%

1/ Each of the above parameters shall be recorded before and after the required burn-in and life tests to determine delta's (Δ).

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4.5.3 High voltage (V_{ZAP}) test of input protection circuits. All input terminals (up to a maximum of four) of the DUT shall be subjected to a voltage pulse from a 100 pF source charged to 400 volts. This destructive test shall be conducted as follows using the test circuit on figure 6:

a. Measure I_{IL} and I_{IH} at the inputs selected, as stated above, at 25°C. The test limit for each input tested shall be ±10 nA at V_{DD} = 18 V dc. Measure I_{SS} on DUT at 25°C. The test limit for this measurement shall be increased a maximum of 20 percent of the specified I_{SS} table III limit at V_{DD} = 18 V dc.



V_{ZAP} = 400 V minimum charge on C1
1 MΩ ≤ R1 ≤ 50 MΩ
R2 = 1.5 kΩ
C1 = 100 pF
S1 = Hg-wetted "bounceless" relay

FIGURE 6. High voltage (V_{ZAP}) test circuit.

b. V_{ZAP} is applied to the DUT in the following modes (see table V) by charging C1 to V_{ZAP} with S1 in position 1 and then switching to position 2.

TABLE V. Modes for high voltage test.

Mode	+ Terminal	- Terminal
1	V_{DD}	Input
2	Input	V_{SS}
3	Input	Associated output

c. Within 24 hours repeat the I_{SS}, I_{IH}, and I_{IL} measurements on the same terminals as performed above. If a DUT exhibits leakage currents in excess of the specified limits after the V_{ZAP} test, it shall be classified as a failure.

4.5.4 Quiescent supply current (I_{SS}) test. When performing quiescent supply current measurements (I_{SS}), the meter shall be placed so that all currents flow through the meter.

4.5.5 Radiation hardness assurance (RHA) testing. The RHA testing shall be performed in accordance with test procedures and sampling specified in table V of method 5005 of MIL-STD-883 and herein:

a. Before irradiation, selected samples shall be assembled in qualified packages and pass the governing electrical parameters (Group A, subgroup 1 at 25°C) and also be subjected to the threshold-voltage test in table VIII in order to calculate the delta threshold (ΔV_T) after irradiation.

b. The devices shall be subjected to a total radiation dose as specified in MIL-M-38510 for the radiation hardness assurance (RHA) level being tested, and meet the end-point electrical parameters as defined in table VI at 25°C, after exposure. The start and completion of the end-point electrical parameter measurements shall not exceed 2 hours following irradiation.

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c. Threshold-voltage test circuit conditions shall be as specified in table VIII and figure 7. In situ and remote testing, the tests shall be performed with the devices biased in accordance with table VII and bias may be interrupted for up to 1 minute to remove devices to the remote bias fixture.

d. After irradiation, the devices shall pass the truth table test as specified in subgroup 7 in table III or if subgroup 7 is not required then an equivalent truth table test shall be performed.

TABLE VI. Radiation hardness end-point electrical parameters at 25°C.

Parameter	All device types	V_{DD}
V_{TN}	0.3 V min	10 V
V_{TP}	2.8 V max	10 V
ΔV_T	1.4 V max	10 V
I_{SS}	100 x max limit	18 V
t_{PLH}	1.35 x max limit	5 V
t_{PHL}	1.35 x max limit	5 V

TABLE VII. Bias during exposure to radiation.

Device type	Pin connections		
	10 V dc through a 30 kΩ to 60 kΩ resistor	V_{SS} = GND	V_{DD} = 10 V dc
01	1,2,5,6,8,9,10,11,12,13	7	14
02	1,2,5,6,8,9,10,11,12,13	7	14
03	1,2,5,6,8,9,12,13	7	14
04	1,2,5,6,8,9,12,13	7	14

NOTE: Pins not designated are open or connected to 10 V dc through a 30 kΩ to 60 kΩ resistor.

4.6 Data reporting. When specified in the contract, a copy of the following data, as applicable, shall be supplied.

a. Attributes data for all screening tests (see 4.2) and variables data for all static burn-in, dynamic burn-in, and steady state life tests.

b. A copy of each radiograph.

c. The quality conformance inspection data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.5).

e. Final electrical parameters data (see 4.2c).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

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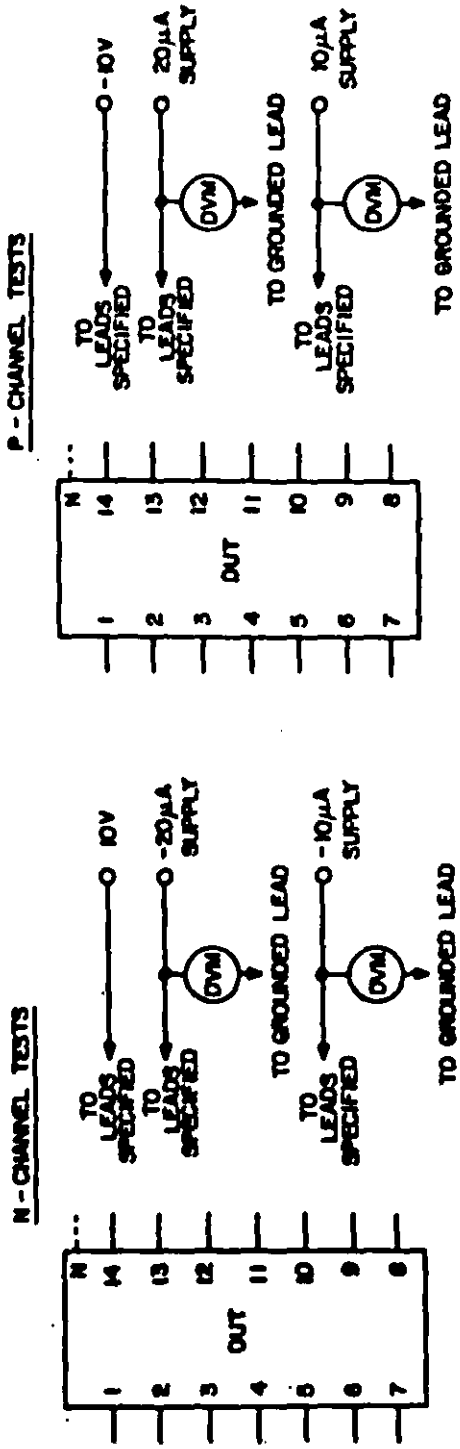


FIGURE 7. Threshold-voltage test circuit.

TABLE VIII. Threshold-voltage test circuit conditions.

Device type	GND	10 V	V_{TN} measured at		GND	-10 V	V_{TP} measured at	
			-20 μA supply	-10 μA supply			20 μA supply	10 μA supply
01	1	2,14		5-13	1	5-13		2,14
02	1	2,14		5-13	1	5-13		2,14
03	1	14		2,5-9,12, 13	1	2,5-9,12, 13		14
04	1	14		2,5-9,12, 13	1	2,5-9,12, 13		14

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6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design application and logisitic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification of qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for quality assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

i. Requirement for total dose radiation testing (see 3.7.1 and 4.5.5), if applicable.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

V_{ZAP} - - - - - - - - - - - - - - - Input test voltage.

C_I - - - - - - - - - - - - - - - Input terminal-to-V_{SS} capacitance.

GND - - - - - - - - - - - - - - - Ground zero voltage potential.

T_A - - - - - - - - - - - - - - - Free air temperature.

V_{IC} - - - - - - - - - - - - - - - Input clamp voltage.

V_{DD} - - - - - - - - - - - - - - - Positive supply voltage.

V_{SS} - - - - - - - - - - - - - - - Negative supply voltage.

I_{SS} - - - - - - - - - - - - - - - Quiescent supply current.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support shall be acquired to device class S for National Aeronautics and Space Administration or class B for Department of Defense (see 1.2.2), lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

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6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	4085B
02	4086B
03	4070B
04	4077B

6.6 Handling. MOS devices must be handled with certain precautions to avoid damage due to accumulation of static charge. Input protective devices have been designed in the chip to minimize the effect of this static build-up. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment and tools.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - MI
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - AR, SM
Navy - AS, OS, CG, MC, SH

Agent:
DLA - ES

Preparing activity:
NASA - NA

(Project 5962-0639-3)

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(See Instructions – Reverse Side)

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MIL-M-38510/172B	MICROCIRCUITS, DIGITAL, CMOS, AND-OR-INVERT, EXCLUSIVE-OR, EXCLUSIVE-NOR GATES, MONOLITHIC SILICON, POSITIVE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
LOGIC	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER (Specify):
b. ADDRESS (Street, City, State, ZIP Code)	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

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